Simulation-Based Practical Evaluation of the Pasqal Neutral-Atom Quantum Computing Platform Using Pulser

Stylianos Mikeos
IRIDA Research Centre for Communication Technologies
University of Cyprus
Supervisor: Professor Dr. Ioannis Krikidis

September 2026

Abstract

This report presents a simulation-based practical evaluation of the Pasqal neutral-atom quantum computing platform using Pulser, Pasqal's Python-based pulse-level programming framework. The work develops a complete workflow from basic neutral-atom control and Rydberg-blockade experiments to reusable graph optimization, communication-inspired MIMO configuration selection, and extended robustness and weighting studies. Small maximum-independent-set (MIS) instances are mapped to atom geometries and checked against exact classical solutions. The communication study compares a QUBO-style formulation, simulated annealing, Pulser-inspired constrained sampling, and a direct no-QUBO formulation based on SNR and Shannon capacity. The extended evaluation then studies controlled dephasing and SPAM sensitivity, fixed-schedule transfer across path graphs from four to eight nodes, and a maximum-weight independent set (MWIS) example using a Detuning Map Modulator (DMM). In the weighted square example, the unique classical optimum `1001` has weight 1.95; its observed probability changes from 0.477 without DMM weighting to 1.000 with DMM weighting, with a 95% Wilson confidence interval of $[0.9962, 1.000]$ for the latter. The results show that Pulser is effective for learning, simulation, and small optimization prototypes, while also demonstrating the importance of pulse tuning, classical validation, uncertainty reporting, and careful separation between simulator results and hardware-level claims.

Keywords:

Pasqal, Pulser, neutral atoms, Rydberg blockade, quantum optimization, maximum independent set, maximum-weight independent set, Detuning Map Modulator, MIMO configuration selection, QUBO, quantum simulation.

Contents

1 Introduction **4**

2 Background on Quantum Computing **4**

3 Pasqal Neutral-Atom Quantum Computing Platform **5**

3.1 Neutral Atoms as Qubits 5

3.2 Rydberg Blockade 5

3.3 Why Pasqal Is Relevant to This Project 6

4 Pasqal Software Ecosystem **6**

4.1 Pulser 6

4.2 Pulser Simulation 6

4.3 Pasqal Cloud and Emulators 6

4.4 Pulser Studio 7

5 Pulser Programming Workflow **7**

6 Beginner Pulser Experiments **7**

6.1 Single-Atom Pulse 8

6.2 Rabi Oscillations 8

6.3 Two-Atom Rydberg Blockade 8

6.4 Three-Atom Chain 8

6.5 Four-Atom Square 8

6.6 Pulse-Parameter Scan 8

7 Reusable Graph-to-Pulser Optimization Pipeline **9**

7.1 Maximum Independent Set Motivation 9

7.2 Pipeline Inputs and Outputs 10

7.3 Square Graph Result 10

7.4 Five-Node Path Graph Result 10

8 MIMO Configuration-Selection Experiment **11**

8.1 Problem Motivation 11

8.2 Small QUBO-Style Setup 11

8.3 Methods Compared 12

8.4 QUBO-Style Results 12

8.5 Pulse Sensitivity and Repeated QUBO Validation 12

9 MIMO Selection Without QUBO **13**

9.1 Direct Communication Objectives 13

9.2 No-QUBO Setup and Results 13

10 Extended Pasqal Evaluation **14**

10.1 Controlled Noise Sensitivity 14

10.2 Fixed-Schedule Path-Graph Scaling 16

10.3 Maximum-Weight Independent Set with a Detuning Map Modulator 17

11 Evaluation and Discussion **18**

12 Limitations **19**

13 Recommendation **20**

14 Future Work **20**

15 Conclusion **21**

A Notebook Workflow Summary **22**

B Example Pseudocode for the Graph-to-Pulser Pipeline **22**

1 Introduction

Quantum computing platforms are becoming increasingly accessible through cloud services and specialized software stacks. However, before a research group can use a particular platform as a research simulation and prototyping tool, it is necessary to understand the platform practically. This includes not only the advertised capabilities, but also the programming workflow, simulation tools, constraints, debugging process, and the type of research problems that naturally fit the device model.

The purpose of this project was to evaluate the Pasqal/Pulser neutral-atom workflow from a practical programming and simulation perspective. Pasqal's technology is based on arrays of neutral atoms controlled by laser pulses. This architecture is especially interesting for analog quantum simulation and optimization-style problems because the physical position of atoms directly affects their interactions. In contrast to a purely gate-based workflow, the Pasqal/Pulser workflow requires the user to think about atom geometries, channels, pulse amplitudes, detunings, phases, durations, and measurement distributions.

The main goal of the work was to build a complete learning pipeline that starts from basic Pulser experiments and gradually moves toward research-relevant simulated optimization studies. The final notebook includes beginner neutral-atom experiments, graph optimization examples, classical validation tools, MIMO configuration-selection experiments, controlled-noise sensitivity tests, graph-size transfer experiments, and a weighted MIS example using local detuning. This report converts that work into a structured technical document that can be used by the research group to understand what was tested in simulation, what was learned, and what should be done next.

The report is organized as follows. Section 2 gives a concise background on quantum computing. Section 3 introduces the Pasqal neutral-atom platform and the role of Rydberg blockade. Section 4 summarizes the relevant software ecosystem. Section 5 explains the Pulser workflow used throughout the project. Sections 6–9 present the foundational experiments and MIMO studies. Section 10 presents the extended noise, scaling, and MWIS/DMM evaluation. Sections 11–13 discuss the findings, limitations, and recommendation. The report ends with future work and conclusions.

2 Background on Quantum Computing

A classical bit stores one of two values, 0 or 1. A quantum bit, or qubit, is a two-level quantum system whose state can be written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \tag{1}$$

where α and β are complex probability amplitudes satisfying

$$|\alpha|^2 + |\beta|^2 = 1. \tag{2}$$

When the qubit is measured in the computational basis, the outcome is 0 with probability $|\alpha|^2$ and 1 with probability $|\beta|^2$.

Quantum computing is not simply the use of quantum mechanics in a device. Modern electronic devices are already governed by quantum mechanics at the microscopic level. Instead, a quantum computer is a machine that deliberately controls and measures quantum states in order to use phenomena such as superposition, interference, and entanglement for information processing. For multiple qubits, the state space grows exponentially with the number of qubits. This is one reason why quantum computers are interesting for simulation and certain algorithmic tasks. [19]

Near-term quantum devices are still limited by noise, imperfect calibration, finite coherence times, limited connectivity, and restricted problem sizes. Therefore, a practical evaluation of a platform must be careful not to overstate results. In this project, the quantum simulations are treated as probabilistic candidate generators. For small problem instances, the results are compared against exact classical solutions in order to avoid ambiguous interpretation.

3 Pasqal Neutral-Atom Quantum Computing Platform

Pasqal develops quantum processors based on neutral atoms. In a neutral-atom quantum computer, individual atoms are trapped and manipulated so that each atom acts as a qubit. The platform can arrange atoms in programmable spatial layouts, and the interactions between atoms depend strongly on their relative distances. This makes the geometry of the atom register an essential part of the program. [13, 14]

3.1 Neutral Atoms as Qubits

The basic idea of a neutral-atom quantum processor is as follows:

- Individual atoms are trapped using optical tweezers;
- Each atom represents a qubit;
- Laser pulses control transitions between atomic states;
- Atoms excited to Rydberg states can interact strongly with nearby atoms;
- The final state is measured and returned as bitstrings.

This architecture is different from many gate-model platforms because the spatial arrangement of the atoms directly affects the physics of the computation. Two atoms that are close to each other interact more strongly than atoms that are far apart. As a result, a problem can sometimes be encoded by choosing an appropriate geometry.

3.2 Rydberg Blockade

Rydberg blockade is one of the key effects used in neutral-atom quantum computing. When an atom is excited to a high-energy Rydberg state, it shifts the energy levels of nearby atoms. If another atom is sufficiently close, this shift can prevent the second atom from being resonantly excited. In practical terms, two atoms inside the blockade region tend not to be excited simultaneously.

This physical effect can be used to encode constraints. If two atoms represent two graph vertices connected by an edge, placing them within the blockade radius discourages the bitstring in which both vertices are selected. This makes Rydberg-atom arrays naturally relevant to independent-set-style optimization problems. The interaction between atoms can be written in simplified form as

$$V_{ij} = \frac{C_6}{R_{ij}^6}, \tag{3}$$

where R_{ij} is the distance between atoms i and j, and C_6 is a coefficient determined by the selected Rydberg level. A common effective Hamiltonian for the driven neutral-atom system is

$$\frac{H}{\hbar} = \sum_i \frac{\Omega(t)}{2}\sigma_i^x - \sum_i \Delta(t) n_i + \sum_{i<j} V_{ij} n_i n_j, \tag{4}$$

Here, $\Omega(t)$ is the Rabi frequency, $\Delta(t)$ is the detuning, σ_i^x is the Pauli-X operator for atom i, and n_i is the Rydberg occupation operator. The interaction term $V_{ij}n_in_j$ contributes only when both atoms i and j are excited. In this project, the details of the Hamiltonian are handled by Pulser; the important practical point is that the user controls the atom positions and pulse parameters.

3.3 Why Pasqal Is Relevant to This Project

Pasqal is relevant to the research group because many engineering problems can be expressed as selection or optimization problems. Examples include graph selection, resource allocation, antenna configuration, and other combinatorial choices. These problems do not automatically become suitable for a neutral-atom processor, but they provide a useful motivation for exploring whether the Pulser workflow can act as a simulated candidate generator for small problem instances.

4 Pasqal Software Ecosystem

The Pasqal ecosystem includes several tools for programming and testing neutral-atom experiments. The most relevant tools for this project are summarized below.

4.1 Pulser

Pulser is an open-source Python package for designing and simulating pulse sequences acting on programmable arrays of neutral atoms [1, 2]. It is the main software tool used in this project. Pulser allows the user to define atom registers, device constraints, pulse sequences, channels, waveforms, measurements, and simulations.

The key reason Pulser was selected is that it exposes the physical programming model of neutral-atom hardware. Instead of only writing high-level gates, the user builds a sequence of physical control pulses. This is useful for learning the platform because it makes the connection between geometry, pulses, interactions, and measured bitstrings explicit.

4.2 Pulser Simulation

Experiments can be simulated before any real-device execution. Simulation is essential for early-stage learning because it allows rapid testing of atom layouts, pulse parameters, and measurement distributions. The experiments in this report are simulation-based, so the results should be interpreted as validation of the programming and modelling workflow, not as a hardware benchmark.

4.3 Pasqal Cloud and Emulators

Pasqal Cloud provides access to cloud-based quantum processing units and emulators [6]. Pasqal documentation describes emulators for simulating neutral-atom quantum systems [7]. These cloud and emulator tools are useful future options, but the present project focuses on local Pulser simulation and reproducible notebook-based experimentation rather than hardware execution; the documented Fresnel workflow provides one route for future QPU job submission [8].

4.4 Pulser Studio

Pulser Studio is a visual, no-code environment for designing neutral-atom pulse sequences [9]. It can be useful for education, visualization, and quick inspection of pulse workflows. However, the main implementation in this project used Pulser directly from Python because the notebook needed programmatic control, parameter sweeps, classical validation, and integration with communication-oriented objective functions.

5 Pulser Programming Workflow

The general workflow used throughout the project is shown in Fig. 1. The same structure appears in the beginner experiments, the graph optimization pipeline, and the MIMO configuration-selection experiment. The workflow follows the register and pulse-sequence programming concepts described in the Pulser documentation [3, 4].

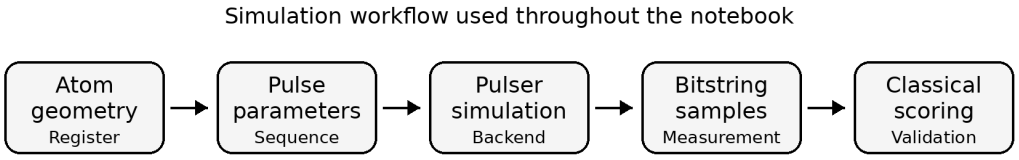


Figure 1: General Pulser workflow used in the project. The workflow starts from atom coordinates and ends with bitstring scoring and interpretation.

The practical steps are:

1. **Define atom coordinates.** The user chooses the spatial positions of the atoms. This step is central because the atom distances determine interaction strengths.
2. **Create the register.** The Pulser register stores the atom positions and assigns labels to the qubits.
3. **Choose the device or emulator.** The selected device defines the physical constraints that the sequence must satisfy, such as allowed channels, pulse durations, and geometry restrictions [5].
4. **Build the pulse sequence.** The sequence defines the time evolution of the experiment using pulses with amplitude, detuning, phase, and duration.
5. **Simulate the sequence.** The simulation computes the expected evolution and measurement distribution.
6. **Sample bitstrings.** Measurement outcomes are returned as bitstrings, where each bit corresponds to an atom state.
7. **Score and analyze results.** Bitstrings are evaluated using the problem objective and compared with a classical benchmark when possible.

A key distinction is that Pulser is closer to a physical pulse-level programming model than a standard gate-based programming model. This gives more control and makes it appropriate for analog neutral-atom workflows, but it also means that pulse-parameter tuning and device and emulator constraints become part of the programming process.

6 Beginner Pulser Experiments

The first part of the project focused on beginner experiments. These were not intended to solve a useful optimization problem immediately. Their purpose was to build intuition about Pulser and the physical behavior of neutral-atom systems.

6.1 Single-Atom Pulse

The first experiment used one atom and a simple pulse. The purpose was to verify that a basic Pulser sequence can be constructed, simulated, and measured. This experiment establishes the minimal workflow: define an atom, declare a channel, add a pulse, run a simulation, and inspect the measurement result.

6.2 Rabi Oscillations

The next experiment scanned pulse duration to observe Rabi oscillations. In a driven two-level system, the population periodically transfers between the ground and excited states. Observing this behavior in simulation is useful because it confirms that the pulse duration and Rabi frequency have the expected effect on the final measurement distribution.

6.3 Two-Atom Rydberg Blockade

The two-atom experiment demonstrates blockade behavior. When two atoms are placed close together, simultaneous excitation is suppressed. Therefore, bitstrings corresponding to both atoms being excited should have lower probability than they would have without interaction. This is the basic physical effect that later supports graph-based optimization.

6.4 Three-Atom Chain

A three-atom chain was used to observe how blockade extends to a small geometry. In the tested chain, the pattern `101` is favored because the two outer atoms can be excited while the middle atom remains unexcited. This demonstrates the connection between atom geometry and allowed excitation patterns.

6.5 Four-Atom Square

A four-atom square was then tested. For a square graph, the expected maximum independent set has size 2, with the selected nodes located at opposite corners. The corresponding checkerboard-like bitstrings are `0110` and `1001`, depending on the ordering convention. This experiment is the bridge between physical blockade behavior and graph optimization.

6.6 Pulse-Parameter Scan

Finally, a pulse-parameter scan was performed. This is important because the measured distribution depends strongly on the selected pulse parameters. In the square-graph scan, the best result obtained in the notebook used a duration of 6000 ns and a final detuning of $8.0\ \mathrm{rad}/\mu\mathrm{s}$, producing a success probability of approximately 0.981.

Table 1: Summary of beginner Pulser experiments.

Experiment	Purpose
Single-atom pulse	Verify the minimal Pulser workflow and one-atom excitation behavior.
Rabi oscillation scan	Show periodic population transfer as pulse duration changes.
Two-atom blockade	Demonstrate suppression of simultaneous nearby excitations.
Three-atom chain	Show how geometry favors patterns such as `101`.
Four-atom square	Connect blockade behavior to checkerboard/independent-set solutions.
Parameter scan	Show that success probability depends strongly on pulse duration and detuning.

The main result of this stage is that the basic Pulser workflow worked correctly and produced the expected qualitative behavior. More importantly, it showed that tuning is not optional. Even for a small four-atom problem, different pulse parameters can produce significantly different output distributions.

7 Reusable Graph-to-Pulser Optimization Pipeline

After the beginner experiments, the next goal was to build a reusable graph-to-Pulser pipeline. The purpose of this pipeline is to take a small graph problem, map it to atom positions, run the Pulser simulation, and compare the sampled bitstrings with the exact classical optimum.

7.1 Maximum Independent Set Motivation

The maximum independent set (MIS) problem asks for the largest set of graph vertices such that no two selected vertices are connected by an edge. It is a natural fit for Rydberg-atom arrays because the Rydberg blockade discourages simultaneous excitation of nearby atoms. If graph edges are represented by atom pairs within the blockade radius, then a valid independent set corresponds to an excitation pattern where adjacent vertices are not simultaneously selected. This connection between Rydberg arrays and MIS has been studied in the literature [10, 11, 12].

For a graph $G = (V, E)$, a binary variable z_i can be assigned to each vertex $i \in V$, where

$$z_i = \begin{cases} 1, & \text{if vertex } i \text{ is selected,} \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

The independent-set constraint is

$$z_i z_j = 0, \quad \forall (i, j) \in E. \tag{6}$$

The objective is to maximize

$$\sum_{i \in V} z_i. \tag{7}$$

7.2 Pipeline Inputs and Outputs

The pipeline takes the following inputs:

- Graph nodes;
- Graph edges;
- Atom coordinates;
- Pulse parameters;
- Number of samples or shots;
- Classical scoring function.

The pipeline produces:

- Pulser measurement samples;
- Probabilities for observed bitstrings;
- Valid and invalid bitstring labels;
- Best observed solution;
- Classical optimum;
- Success probability.

The most important design choice is that every quantum-inspired result is compared with a classical benchmark. This makes the interpretation clear: the simulation is not treated as automatically correct; it is evaluated against an exact reference for small instances.

7.3 Square Graph Result

The first graph test was the square graph. For the square graph, the maximum independent-set size is 2. The two optimal bitstrings are `0110` and `1001`, corresponding to selecting opposite corners of the square.

In the saved final run, the untuned reusable-pipeline simulation assigned probabilities 0.369 and 0.354 to `1001` and `0110`, respectively, giving a total success probability of 0.723. A pulse-parameter scan then tested durations from 2000 to 6000 ns and final detunings from 4 to $10\ \mathrm{rad}/\mu\mathrm{s}$. The best tested setting was a duration of 6000 ns and a final detuning of $8.0\ \mathrm{rad}/\mu\mathrm{s}$, for which the observed success probability was 0.991 over 1000 samples.

This result demonstrates that the graph geometry can encode the intended MIS constraint structure, but the probability of measuring an exact optimum remains strongly dependent on the pulse schedule.

7.4 Five-Node Path Graph Result

The same reusable pipeline was tested on a five-node path graph. The exact classical maximum independent set has size 3 and is represented by the bitstring `10101`. In the saved final run, `10101` was the most frequent sampled state with probability 0.562 over 1000 samples.

The result reinforces the probabilistic interpretation of the workflow. The exact optimum is favoured by the simulated dynamics, but its sampling probability depends on the instance, geometry, pulse schedule, and finite sampling process.

Table 2: Graph optimization results from the reusable graph-to-Pulser pipeline in the final executed notebook.

Graph	Classical result	Optimal bitstrings	Pulser result
Square graph	MIS size 2	`0110`, `1001`	Success probability 0.723 before tuning and 0.991 at the best tested pulse setting.
5-node path	MIS size 3	`10101`	The exact optimum was the most likely result, with success probability 0.562.

8 MIMO Configuration-Selection Experiment

The communication-oriented part of the project connects the Pasqal workflow to wireless communications. The selected example is a MIMO configuration-selection problem motivated by the reconfigurable-antenna formulation studied in [15]. The purpose is not to claim a large-scale quantum advantage, but to test whether a communications-related selection problem can be expressed in a quantum-optimization style and validated against a classical benchmark.

8.1 Problem Motivation

In wireless communication systems, performance depends on configuration choices. For example, different antenna states, transmit/receive configurations, or reconfigurable elements can produce different effective channel qualities. The problem considered here is a simplified version of this idea: choose one state per antenna so that the resulting communication metric is maximized.

This type of problem is naturally combinatorial. If each antenna has multiple possible states, the number of configurations grows rapidly. For small cases, exhaustive search is possible and gives an exact benchmark. For larger cases, candidate-generation and heuristic optimization methods become more interesting.

8.2 Small QUBO-Style Setup

The first MIMO experiment used:

- Transmit antennas: $N_T = 2$;
- Receive antennas: $N_R = 2$;
- States per antenna: $N = 2$;
- Eight binary variables;
- Sixteen valid configurations.

The binary vector b encodes which state is selected. The QUBO-style objective was written as

$$\max_{b} \; b^T Q b, \tag{8}$$

subject to the constraint that exactly one state is selected for each antenna group. In practice, QUBO formulations usually include penalty terms to discourage invalid selections. This makes them compatible with binary optimization workflows, but it also introduces a tuning problem because penalty strengths can affect the result.

8.3 Methods Compared

Three methods were compared:

1. **Exhaustive search:** Checks every valid configuration and gives the exact optimum.
2. **QUBO-style simulated annealing:** Tests whether the QUBO formulation can recover the same optimum.
3. **Pulser-inspired neutral-atom sampling:** Produces candidate bitstrings that are then scored classically.

The exhaustive search is the ground truth. The other methods are evaluated by whether they can find the same best feasible bitstring.

8.4 QUBO-Style Results

For the representative 8-variable MIMO instance, exhaustive search and simulated annealing identified the same optimum:

- Exhaustive-search optimum: `10101010`;
- Simulated-annealing best: `10101010`;
- Simulated annealing matched exhaustive search: True;
- Pulser-inspired feasible probability: 0.764;
- Pulser-inspired optimal probability: 0.036;
- Best observed feasible bitstring: `10101010`;
- Best observed feasible bitstring matched the global optimum: True.

The best solution was therefore present in the Pulser-generated candidate set, but it was not the most frequent sample. This is consistent with the role of the neutral-atom prototype in this study: the geometry encodes the one-hot feasibility structure, while the complete weighted MIMO objective is evaluated classically after sampling.

Table 3: QUBO-style MIMO configuration-selection result for the representative final run.

Quantity	Result
Exhaustive-search optimum	`10101010`
Simulated annealing best	`10101010`
Matches exhaustive optimum	True
Pulser-inspired feasible probability	0.764
Pulser-inspired optimal probability	0.036
Best observed feasible bitstring	`10101010`
Best observed feasible bitstring matches optimum	True

8.5 Pulse Sensitivity and Repeated QUBO Validation

The Pulser-inspired MIMO prototype was also evaluated over a small grid of pulse durations and final detunings. The best tested scan point used a duration of 4000 ns and final detuning of $10\ \mathrm{rad}/\mu\mathrm{s}$, giving an observed feasible probability of 0.946 and an optimal-bitstring probability of 0.068. The fact that these metrics vary across the scan reinforces that candidate quality depends on the pulse schedule even when the atom layout is fixed.

Separately, the QUBO simulated-annealing solver was checked over ten independently generated channel realizations. For each realization, exhaustive search supplied the exact reference solution and repeated simulated-annealing runs were used to retain the best candidate. The best-of-runs hit rate was 1.0 across the ten tested cases. This result validates the small QUBO implementation for the tested instances; it does not imply that every stochastic annealing run reaches the optimum.

9 MIMO Selection Without QUBO

A second MIMO version was added without using a QUBO formulation. The motivation was practical clarity. While QUBO is useful for connecting problems to binary optimization frameworks, it introduces penalties and can obscure the original communication objective. The no-QUBO version directly evaluates the communication metric for each configuration.

9.1 Direct Communication Objectives

Two objective functions were considered:

- Signal-to-noise ratio (SNR);
- Shannon capacity.

A common capacity expression for a MIMO channel is

$$C = \log_2 \det \left(I_{N_R} + \frac{\rho}{N_T} H H^{\dagger} \right), \tag{9}$$

where H is the channel matrix, $H^{\dagger}$ is its Hermitian transpose, ρ is the SNR parameter, N_T is the number of transmit antennas, and N_R is the number of receive antennas. The exact implementation in the notebook evaluates candidate configurations and compares the best selected configuration with exhaustive search.

9.2 No-QUBO Setup and Results

The no-QUBO experiment used:

- Transmit antennas: $N_T = 2$;
- Receive antennas: $N_R = 2$;
- States per antenna: $N = 3$;
- Eighty-one possible configurations.

For the representative final-run channel instance, the exhaustive-search and multi-start simulated-annealing solutions matched for both objectives. The optimal SNR score was 5.5762, while the optimal capacity was 7.3183. The SNR-optimal and capacity-optimal configurations were not identical, illustrating that the two communication metrics need not favour the same antenna-state selection.

The experiment was then repeated over ten independently generated channel realizations. The best-of-runs SNR hit rate and capacity hit rate were both 1.0 relative to exhaustive search.

Table 4: No-QUBO MIMO configuration-selection result.

Quantity	Result
Transmit antennas	$N_T = 2$
Receive antennas	$N_R = 2$
States per antenna	$N = 3$
Configurations	81
Representative SNR optimum	5.5762
Representative capacity optimum	7.3183
SNR optimum matched exhaustive search	True
Capacity optimum matched exhaustive search	True
10-instance SNR best-of-runs hit rate	1.0
10-instance capacity best-of-runs hit rate	1.0

This direct version is easier to explain to a communications audience because it retains the original SNR and capacity objectives without QUBO penalty terms. Its main limitation is that it is not, by itself, a direct neutral-atom Hamiltonian mapping. It serves as a classical communication-objective benchmark and as a comparison point for the Pulser-inspired candidate-generation workflow.

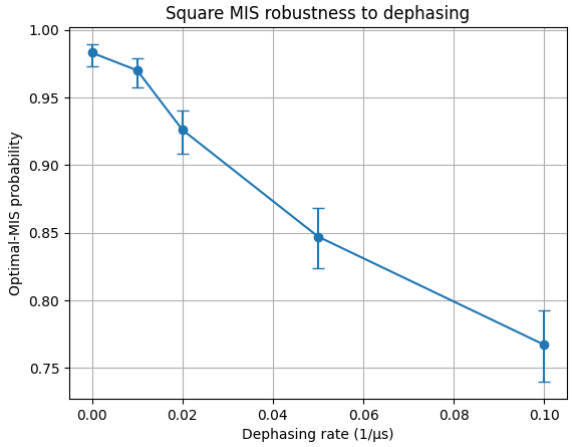
10 Extended Pasqal Evaluation

After the initial graph and MIMO studies, the notebook was extended in three directions: controlled noise sensitivity, graph-size transfer, and weighted independent-set optimization using local detuning. These experiments use the same principle as the earlier sections: the simulator output is treated probabilistically and is checked against an exact classical reference whenever the problem size permits.

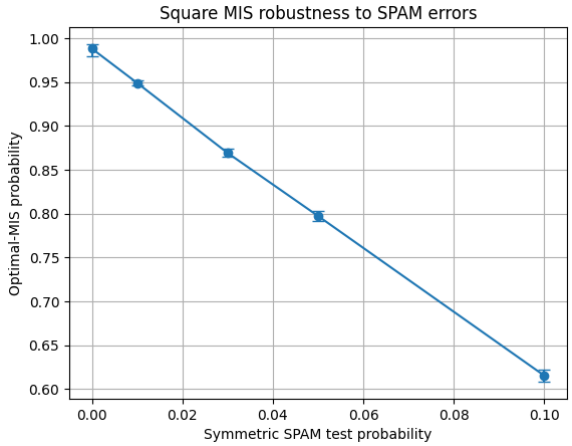
10.1 Controlled Noise Sensitivity

Pulser provides a `NoiseModel` for emulator studies, including dephasing and state-preparation-and-measurement (SPAM) parameters [16]. The tuned four-node square sequence was held fixed while controlled noise parameters were varied. These values are sensitivity variables used inside the simulator; they are not presented as measured parameters of a specific Pasqal QPU.

For the dephasing study, the Rydberg dephasing rate was varied from 0 to $0.10\ \mu\mathrm{s}^{-1}$. The observed probability of sampling one of the exact square-MIS solutions decreased from 0.983 to 0.767. A second symmetric SPAM test assigned the same sweep value to state-preparation error, false-positive readout probability, and false-negative readout probability. Across the tested range from 0 to 0.10, the observed optimal-solution probability decreased from 0.988 to 0.6153.



(a) Controlled Rydberg-dephasing sweep.



(b) Symmetric SPAM sensitivity sweep.

Figure 2: Noise sensitivity of the tuned square-graph MIS sequence. Error bars show 95% Wilson confidence intervals for the sampled probabilities.

Table 5: Controlled-noise sensitivity results for the tuned square graph.

Study	**Level**	$\mathbf{P}(\text{optimal})$	**95% Wilson CI**
Dephasing	0.00	0.983	$[0.9729, 0.9894]$
Dephasing	0.01	0.970	$[0.9575, 0.9789]$
Dephasing	0.02	0.926	$[0.9081, 0.9406]$
Dephasing	0.05	0.847	$[0.8234, 0.8680]$
Dephasing	0.10	0.767	$[0.7398, 0.7921]$
Symmetric SPAM	0.00	0.9880	$[0.9791, 0.9931]$
Symmetric SPAM	0.01	0.9489	$[0.9458, 0.9519]$
Symmetric SPAM	0.03	0.8691	$[0.8644, 0.8737]$
Symmetric SPAM	0.05	0.7973	$[0.7916, 0.8028]$
Symmetric SPAM	0.10	0.6153	$[0.6086, 0.6221]$

The SPAM simulation aggregates samples over multiple stochastic trajectories, so its effective sample count differs from the simple 1000-shot cases used elsewhere in the notebook. The interval widths should therefore not be compared directly across experiments as though all probabilities were estimated from the same number of samples. The main conclusion is qualitative and robust: increasing the controlled noise level reduces the probability of measuring an exact MIS solution.

10.2 Fixed-Schedule Path-Graph Scaling

To test whether a single pulse schedule transfers across problem sizes, path graphs P_4 through P_8 were simulated using the same sequence parameters: 3000 ns duration, 6 $\mathrm{rad}/\mu\mathrm{s}$ final detuning, and 2.5 $\mathrm{rad}/\mu\mathrm{s}$ maximum amplitude. No per-size retuning was performed.

Table 6: Fixed-schedule path-graph scaling results.

N	**Exact MIS size**	$\mathbf{P}(\text{optimal})$	$\mathbf{P}(\text{valid})$	$\mathbf{E}[\lvert\mathbf{S}\rvert]$	**Best valid**
4	2	0.849	0.999	1.851	2
5	3	0.576	1.000	2.477	3
6	3	0.769	1.000	2.732	3
7	4	0.373	1.000	3.187	4
8	4	0.683	1.000	3.606	4

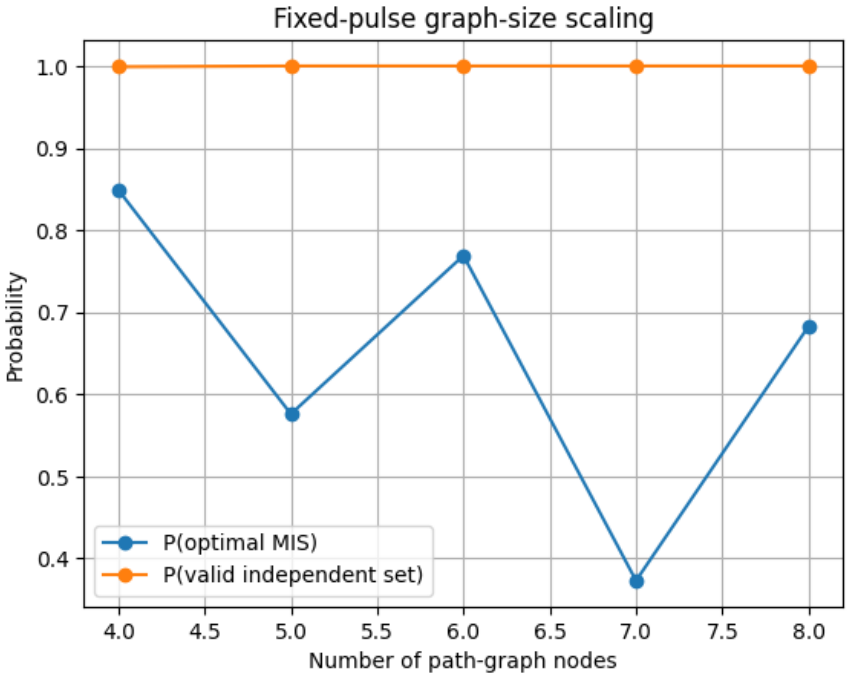


Figure 3: Optimal-MIS and valid-independent-set probabilities for path graphs P_4–P_8 under one fixed pulse schedule.

The valid-independent-set probability remained between 0.999 and 1.000 for all tested graph sizes, and the exact MIS cardinality was observed in every case. The probability of sampling an exact optimum was non-monotonic, ranging from 0.373 to 0.849. This should be interpreted as a fixed-schedule transfer study, not as an asymptotic scaling claim. Different graph sizes have different spectra, degeneracies, and adiabatic requirements, so a separate pulse optimization for each N could give different results. Local QuTiP runtime was also recorded in the notebook, but it is not treated as a QPU runtime benchmark.

10.3 Maximum-Weight Independent Set with a Detuning Map Modulator

The final extension moves from unweighted MIS to maximum-weight independent set (MWIS). For positive vertex weights w_i, the problem is

$$\max_{x_i \in \{0,1\}} \sum_i w_i x_i, \tag{10}$$

subject to

$$x_i + x_j \leq 1, \qquad \forall (i,j) \in E. \tag{11}$$

Pulser supports local detuning through a Detuning Map Modulator (DMM), where site-dependent weights determine the fraction of a DMM detuning waveform applied to each atom [17]. Pasqal's MWIS tutorial uses this mechanism to favour high-weight vertices [18]. In the notebook, the four-node square was assigned weights

$$[w_0, w_1, w_2, w_3] = [1.00, 0.35, 0.45, 0.95]. \tag{12}$$

The exact classical weighted optimum is `1001`, with total weight

$$1.00 + 0.95 = 1.95. \tag{13}$$

The simulation used `MockDevice` and a QAA-style global pulse with a 40 μs duration. The maximum Rabi frequency and final detuning were set to $145.216\ \mathrm{rad}/\mu\mathrm{s}$ according to the interaction-based heuristic used in the MWIS prototype. A direct final-Hamiltonian energy check identified `1001` as the lowest-energy computational-basis candidate before sampling. Because `MockDevice` is a virtual test device, these pulse parameters are used to demonstrate the encoding and are not presented as a hardware-feasible schedule for a specific Pasqal QPU.

Without DMM weighting, the square remains symmetric between its two diagonal MIS solutions. In the saved run, 523 of 1000 samples were `0110` and 477 were `1001`, so the particular bitstring that later becomes the weighted optimum had probability 0.477. After enabling the DMM, all 1000 sampled bitstrings were `1001`. The observed weighted-optimum probability was therefore 1.000, with a 95% Wilson confidence interval of $[0.9962, 1.000]$.

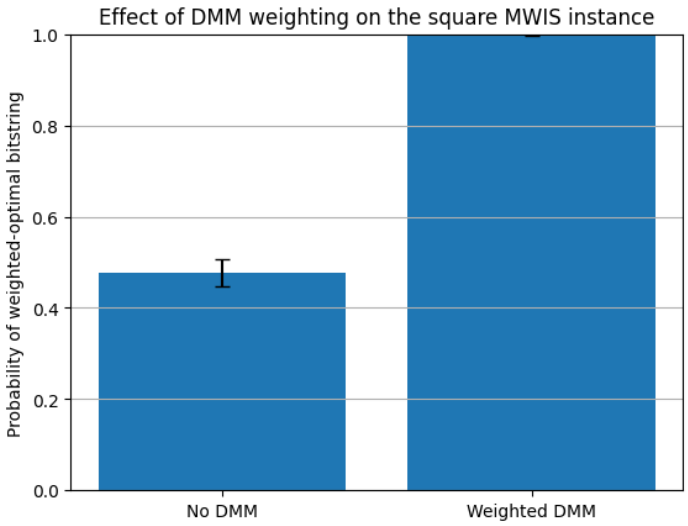


Figure 4: Probability of the classically verified weighted-optimal bitstring `1001` without and with DMM weighting. Error bars show 95% Wilson confidence intervals.

Table 7: Weighted square-graph MWIS result.

Case	$\mathbf{P}(\texttt{1001})$	**95% Wilson CI**	**Dominant outcomes**
No DMM	0.477	$[0.4462, 0.5080]$	`0110`: 523; `1001`: 477
Weighted DMM	1.000	$[0.9962, 1.000]$	`1001`: 1000

This experiment is a stronger neutral-atom-specific optimization result than the earlier MIMO prototype because the linear vertex weights are introduced directly through the local-detuning term rather than only in classical post-processing. The result remains a small idealized simulation and should not be interpreted as evidence that an arbitrary quadratic MIMO objective can be represented exactly by the same DMM construction.

11 Evaluation and Discussion

The final notebook combines several types of validation. Exact classical search is used for all small graph and configuration-selection problems where enumeration is practical. Stochastic quantum or heuristic outputs are then evaluated using metrics appropriate to their role: optimal-solution probability, valid-solution probability, best-observed objective value, confidence intervals, and best-of-runs agreement with the exact benchmark.

The graph experiments show two distinct behaviours. First, a suitable neutral-atom geometry can strongly suppress invalid independent-set configurations. Second, the probability of sampling an exact optimum is sensitive to the pulse schedule and problem instance. This is visible in the tuned square result, the five-node path result, and the non-monotonic P_4–P_8 fixed-schedule study. The noise sweeps add a third observation: even when the noiseless schedule performs well, controlled dephasing and SPAM errors reduce the measured optimal-solution probability.

The weighted-MIS result demonstrates an additional capability of the neutral-atom model. Geometry represents the conflict constraints, while DMM-based local detuning breaks the degeneracy between otherwise equivalent independent sets according to a linear vertex-weight objective. In the tested square, the no-DMM sequence splits its probability approximately equally between the two diagonal MIS states, whereas the weighted sequence concentrates all 1000 saved samples on the classically verified MWIS solution `1001`. This is an encoding result for a small simulator instance, not a hardware-performance benchmark.

The MIMO experiments must be interpreted differently. The QUBO simulated-annealing and direct no-QUBO solvers can be validated exactly on the small tested channel instances, and both recover the exhaustive-search solution when the best result over repeated runs is retained. The Pulser MIMO prototype, however, does not directly encode the full weighted matrix Q in the neutral-atom Hamiltonian. Its appropriate interpretation is therefore constrained candidate generation followed by classical objective evaluation.

Taken together, the experiments support Pulser as a useful research-development environment for exploring which optimization structures map naturally to neutral atoms. MIS-type conflict constraints and linear vertex weighting have a direct physical interpretation through blockade geometry and local detuning. More general communication objectives require additional modelling work before they can be described as direct neutral-atom optimization mappings.

12 Limitations

This project remains a simulation-based practical evaluation rather than a QPU benchmark. The main limitations are:

- **Simulation-based results:** All quantum results were produced with Pulser-compatible simulators or virtual devices and should be interpreted as workflow and encoding studies.
- **Small problem sizes:** The graph and MIMO instances remain small enough for exact classical verification.
- **Parameter dependence:** Optimal-solution probability depends strongly on pulse duration, detuning, amplitude, and the selected graph geometry.
- **Controlled rather than calibrated noise:** The dephasing and SPAM sweeps are simulator sensitivity tests, not measurements of a specific Pasqal QPU noise process.
- **Virtual-device MWIS demonstration:** The DMM experiment uses `MockDevice`. Its 40 μs schedule and pulse scales demonstrate the weighted encoding but are not claimed to satisfy a specific hardware device's execution limits.
- **Finite stochastic sampling:** Reported probabilities vary between executions. Confidence intervals quantify sampling uncertainty for selected experiments, but the project does not yet provide large ensembles over random graph instances or pulse seeds.
- **QUBO penalty and model sensitivity:** QUBO-style formulations depend on penalty design, and the Pulser MIMO prototype does not directly realize the complete weighted MIMO objective.
- **Local simulator runtime is not QPU runtime:** The measured path-graph simulation times reflect local QuTiP computation and should not be used to infer neutral-atom hardware scaling.
- **No claim of quantum advantage:** The tested cases are too small and too simulation-focused to support claims of speedup or computational advantage.

The central practical limitation is therefore not whether the simulator can produce the correct answer, but how reliably a chosen physical encoding and pulse schedule concentrates probability on useful solutions under realistic device constraints and noise.

13 Recommendation

Based on the completed experiments, Pulser is worth retaining as a simulation and prototyping environment for the group. The project now goes beyond basic platform familiarization: it includes exact classical validation, pulse scans, repeated random-channel tests, statistical uncertainty, controlled-noise sensitivity, graph-size transfer, and a direct local-detuning MWIS demonstration.

The next phase should therefore focus less on adding isolated examples and more on increasing realism and reproducibility:

1. Reproduce a small subset of the final notebook on a device-constrained Pasqal emulator and, if access becomes available, on a QPU;
2. Distinguish clearly between schedules that are valid on a specific hardware device and schedules intended only for `MockDevice` or unconstrained simulation;
3. Repeat the graph experiments over random graph families and multiple stochastic seeds, not only path and square geometries;
4. Compare fixed schedules with per-instance or per-size pulse optimization;
5. Use hardware-informed noise parameters when a specific target device is selected;
6. Extend the communication study to larger and more realistic channel models while preserving exact classical validation on tractable subinstances;
7. Investigate which parts of communication objectives can be mapped directly to neutral-atom interactions or local detunings, and which parts still require classical post-processing.

For an initial hardware-oriented benchmark, the four-node square remains the clearest starting point because its exact solutions are known, its geometry is transparent, and its pulse schedule has already been characterized in detail. The weighted square can then be used as a second benchmark if a target device supports an appropriate local-detuning mechanism.

14 Future Work

Future work can extend the current study in four main directions. First, the graph benchmark set should be broadened beyond paths and a single square to include random unit-disk graphs, multiple graph densities, and instances large enough to expose simulator limits while still allowing classical verification where possible. Second, pulse optimization should be made more systematic by comparing grid search with automated optimization methods and by reporting performance across multiple stochastic runs.

Third, the noise study should move from controlled sensitivity variables toward device-informed models. This includes separating state-preparation, readout, dephasing, and relaxation contributions rather than varying several SPAM terms together, and comparing emulator predictions with QPU data if hardware access becomes available. Fourth, the communication direction should investigate mappings in which more of the objective is represented physically rather than scored after measurement. The DMM experiment shows that linear vertex weights can be introduced directly through site-dependent detuning; the next research question is how far this idea can be extended before the required objective no longer matches the neutral-atom Hamiltonian structure.

For MIMO configuration selection, larger antenna/state spaces, correlated and measured channel models, imperfect channel-state information, and practical reconfiguration constraints are natural extensions. The existing exact-search and simulated-annealing implementations provide useful validation tools for small subinstances while these more realistic models are developed.

15 Conclusion

This project developed and extended a practical simulation workflow for Pasqal neutral-atom programming using Pulser. The study began with pulse-level experiments such as single-atom control, Rabi oscillations, and Rydberg blockade, progressed to a reusable graph-to-Pulser MIS pipeline, and then explored communication-inspired MIMO configuration selection with both QUBO and direct objective formulations.

The extended evaluation adds three results that strengthen the original study. Controlled dephasing and SPAM sweeps quantify how the tuned square-MIS probability degrades under simulator noise. A fixed-schedule path-graph study shows that valid independent sets remain dominant from four to eight nodes while exact-optimum probability varies with problem size. Finally, the DMM-based MWIS experiment demonstrates direct linear weighting in the neutral-atom Hamiltonian: the unique classical solution `1001`, with weight 1.95, appears with probability 0.477 in the symmetric no-DMM reference and with observed probability 1.000 after DMM weighting, with a 95% Wilson interval of $[0.9962, 1.000]$.

The principal findings are therefore:

- Pulser provides a practical pulse-level environment for learning and prototyping neutral-atom experiments;
- Atom geometry and Rydberg blockade naturally represent independent-set-style conflict constraints;
- Pulse parameters materially affect the probability of measuring an exact optimum;
- Controlled simulator noise reduces optimization success probability and should be included in robustness studies;
- Local detuning through DMM can encode linear vertex weights and resolve degeneracy between geometric MIS solutions;
- MIMO configuration selection can be connected to the workflow, but the present Pulser MIMO prototype still evaluates the full communication objective classically;
- Exact classical baselines and uncertainty reporting are essential for interpreting small stochastic quantum-simulation results;
- The current work does not demonstrate quantum advantage or hardware-level performance.

Overall, Pasqal/Pulser is a useful research simulation and prototyping environment for the group, particularly when the problem structure matches neutral-atom interactions. The final notebook and this report provide a reproducible foundation for a future transition from idealized local simulation toward device-constrained emulation, hardware-informed noise studies, and eventually QPU experiments if access is available.

A Notebook Workflow Summary

The final notebook follows this approximate structure:

1. Import Pulser, simulation, numerical, plotting, and optimization libraries;
2. Define helper functions for simulation, graph validation, probability reporting, and bitstring scoring;
3. Run beginner single-atom, Rabi, blockade, chain, square, and pulse-scan experiments;
4. Define the reusable graph-to-register and graph-scoring utilities;
5. Validate square and path MIS instances against exact classical search;
6. Define the QUBO-style MIMO configuration-selection problem and compare exhaustive search, simulated annealing, and Pulser-inspired sampling;
7. Validate the QUBO solver over multiple random channel realizations;
8. Evaluate the direct no-QUBO MIMO formulation using SNR and capacity objectives and repeated channel tests;
9. Evaluate controlled dephasing and SPAM sensitivity with confidence intervals;
10. Evaluate one fixed pulse schedule across path graphs P_4–P_8;
11. Construct and validate a weighted square-graph MWIS experiment using DMM local detuning;
12. Summarize results, limitations, and interpretation.

B Example Pseudocode for the Graph-to-Pulser Pipeline

Listing 1: High-level pseudocode for the reusable graph-to-Pulser pipeline.

```
def run_graph_pulser_experiment(graph, coordinates, pulse_params, shots):

    # 1. Classical reference

    classical_optimum = solve_maximum_independent_set_exact(graph)

    # 2. Build atom register from graph coordinates

    register = Register.from_coordinates(coordinates)

    # 3. Build Pulser pulse sequence

    sequence = Sequence(register, device)

    sequence.declare_channel("rydberg", "rydberg_global")

    sequence.add(make_pulse(pulse_params), "rydberg")

    sequence.measure("ground-rydberg")

    # 4. Simulate and sample bitstrings

    result = simulate(sequence)

    samples = result.sample_final_state(shots)

    # 5. Score samples

    scored = score_bitstrings(samples, graph)

    success_probability = probability_of_optimum(scored, classical_optimum)

    return scored, classical_optimum, success_probability
```

References

[1] H. Silvério, S. Grijalva, C. Dalyac, L. Leclerc, P. J. Karalekas, N. Shammah, M. Beji, L.-P. Henry, and L. Henriet, "Pulser: An open-source package for the design of pulse sequences in programmable neutral-atom arrays," *Quantum*, vol. 6, p. 629, Jan. 2022. [Online]. Available: https://quantum-journal.org/papers/q-2022-01-24-629/

[2] Pasqal, "Pulser documentation," accessed September 2026. [Online]. Available: https://docs.pasqal.com/pulser/

[3] Pasqal, "Programming a neutral-atom QPU," accessed September 2026. [Online]. Available: https://docs.pasqal.com/pulser/programming/

[4] Pasqal, "Register and Rydberg-atom interactions," accessed September 2026. [Online]. Available: https://docs.pasqal.com/pulser/register/

[5] Pasqal, "Hardware specifications," accessed September 2026. [Online]. Available: https://docs.pasqal.com/pulser/hardware/

[6] Pasqal, "Welcome to Pasqal Cloud Services," accessed September 2026. [Online]. Available: https://docs.pasqal.com/cloud/

[7] Pasqal, "Pasqal Cloud emulators," accessed September 2026. [Online]. Available: https://docs.pasqal.com/cloud/emulators/

[8] Pasqal, "Send a job to Fresnel," accessed September 2026. [Online]. Available: https://docs.pasqal.com/cloud/fresnel-job/

[9] Pasqal, "Pulser Studio," accessed September 2026. [Online]. Available: https://docs.pasqal.com/pulserstudio/

[10] H. Pichler, S.-T. Wang, L. Zhou, S. Choi, and M. D. Lukin, "Quantum optimization for maximum independent set using Rydberg atom arrays," arXiv:1808.10816, 2018. [Online]. Available: https://arxiv.org/abs/1808.10816

[11] S. Ebadi *et al.*, "Quantum optimization of maximum independent set using Rydberg atom arrays," *Science*, vol. 376, no. 6598, pp. 1209–1215, 2022. [Online]. Available: https://www.science.org/doi/10.1126/science.abo6587

[12] C. Dalyac, L.-P. Henry, M. Kim, J. Ahn, and L. Henriet, "Exploring the impact of graph locality for the resolution of the maximum-independent-set problem with neutral atom devices," *Physical Review A*, vol. 108, p. 052423, 2023. [Online]. Available: https://doi.org/10.1103/PhysRevA.108.052423

[13] L. Henriet, L. Béguin, A. Signoles, T. Lahaye, A. Browaeys, G.-O. Reymond, and C. Jurczak, "Quantum computing with neutral atoms," *Quantum*, vol. 4, p. 327, Sept. 2020. [Online]. Available: https://quantum-journal.org/papers/q-2020-09-21-327/

[14] T. Macrì, "Quantum Computing with Neutral Atoms," in *Quantum Technologies*, J. Jang-Jaccard *et al.*, Eds. Cham, Switzerland: Springer, 2026, ch. 3, pp. 25–31. [Online]. Available: https://doi.org/10.1007/978-3-031-90727-2_3

[15] I. Krikidis, C. Psomas, A. K. Singh, and K. Jamieson, "Optimizing configuration selection in reconfigurable-antenna MIMO systems: Physics-inspired heuristic solvers," *IEEE Transactions on Communications*, vol. 72, no. 12, pp. 8010–8023, 2024. [Online]. Available: https://doi.org/10.1109/TCOMM.2024.3420768

[16] Pasqal, "Noise Model and Noisy Simulations," accessed September 2026. [Online]. Available: https://docs.pasqal.com/pulser/noise_model/

[17] Pasqal, "Local Addressability with DMM," accessed September 2026. [Online]. Available: https://docs.pasqal.com/pulser/tutorials/dmm/

[18] Pasqal, "QAA to solve a MWIS problem," accessed September 2026. [Online]. Available: https://docs.pasqal.com/pulser/tutorials/mwis/

[19] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th anniversary ed. Cambridge, U.K.: Cambridge University Press, 2010.